ModelIt!

Teacher's Guide

Is Fast Fashion Destroying the Planet?

Modeling the Environmental Life Cycle of Clothing

Quick Reference

Grade Level	Grade 11L1
NGSS Standard	HS-ESS3-2, HS-ETS1-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS3-2, HS-ETS1-3

Evaluate competing design solutions for developing, managing, and utilizing energy and mineral resources based on cost-benefit ratios. Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints, including cost, safety, reliability, and aesthetics, as well as possible social, cultural, and environmental impacts.

Three-Dimensional Learning

Science Practice	**Analyzing and Interpreting Data** — Students analyze life cycle assessment data to compare the environmental impact of different materials, production volumes, and garment lifespans, interpreting complex trade-offs across multiple environmental metrics.
Disciplinary Core Idea	**ESS3.C Human Impacts on Earth Systems / ETS1.B Developing Possible Solutions** — The fashion industry significantly impacts Earth's water systems, atmosphere, and land through resource extraction, manufacturing pollution, and waste. Design solutions must balance environmental, economic, and social constraints.
Crosscutting Concept	**Energy and Matter** — Students trace the flow of matter (fibers, water, chemicals) and energy (fossil fuels for production and transport) through the complete life cycle of a garment, identifying where waste and pollution enter the environment at each stage.

CAST Testing Alignment

MULTIPLE CHOICE:

Consumer A buys 50 polyester garments per year and wears each an average of 7 times. Consumer B buys 15 organic cotton garments per year and wears each an average of 80 times. Based on your model, which consumer has the lower total annual environmental footprint, and approximately by what factor?

CONSTRUCTED RESPONSE:

Using your model, explain why garment lifespan is a more powerful lever for reducing fashion's environmental impact than material choice alone. Include specific data from your simulations comparing the impact of doubling lifespan versus switching to sustainable materials.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The fashion industry has undergone a radical transformation over the past two decades. Where previous generations bought fewer, higher-quality garments and wore them for years, the rise of fast fashion has created a system where the average garment is worn just 7-10 times before being discarded. The industry now produces over 100 billion garments annually — more than 12 for every person on Earth — with devastating environmental consequences that are largely invisible to consumers.

The Scale of Fast Fashion

The global fashion industry produces over 100 billion garments per year, generating $1.7 trillion in revenue. The average consumer buys 60% more clothing than 15 years ago but keeps each item half as long. Brands like Shein add 6,000 new styles per DAY to their website. This volume is driven by a business model that profits from disposability: cheap materials, minimal construction quality, and aggressive trend cycling that makes last month's styles feel obsolete. The result is a volume of production that no waste system can handle.

Water: The Hidden Cost of Cotton and Dye

Cotton accounts for 24% of global fiber production and is extraordinarily water-intensive: growing enough cotton for a single t-shirt requires approximately 2,700 liters of water — equivalent to what one person drinks in 2.5 years. The Aral Sea in Central Asia has shrunk by 90% largely due to cotton irrigation. Textile dyeing is the second-largest water polluter globally, using over 1.5 trillion liters of water annually and discharging heavy metals, formaldehyde, and other toxins into rivers. In countries like Bangladesh, rivers near textile factories run the color of the season's trends.

Carbon and Microplastics: The Polyester Problem

Polyester, made from petroleum, now accounts for 60% of all clothing fiber. Its production releases approximately 700 million tons of CO2 annually. But the carbon footprint is only part of the problem. When polyester garments are washed, they shed up to 700,000 microplastic fibers per load. These fibers pass through wastewater treatment plants, enter oceans and rivers, are consumed by marine life, and bioaccumulate up the food chain. Microplastics have been found in human blood, lung tissue, and placenta. Unlike organic fibers, synthetic microplastics persist in the environment for hundreds of years.

The Waste Crisis: Where Clothes Go to Die

Of the 92 million tons of textile waste produced annually, 87% goes to landfills or is incinerated. Only 13% is recycled, and of that, most is 'downcycled' into lower-value products like insulation or rags rather than new clothing. The fundamental problem is that most fast fashion is made from fiber blends (polyester-cotton, nylon-elastane) that current recycling technology cannot separate. A polyester shirt in a landfill will take 200+ years to decompose, leaching chemicals throughout. The fashion industry's linear model — extract, produce, consume, discard — is fundamentally incompatible with environmental sustainability.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify production volume, material type, garment lifespan, water consumption, carbon emissions, and textile waste generated as the key components of the fashion environmental impact system.

E **Establish Relationships**

Students determine that production volume, material type, and garment lifespan are external factors driven by industry and consumer choices, while water consumption, carbon emissions, and textile waste are environmental outcomes determined by the interaction of production decisions with material properties and disposal methods.

V **Visualize & Model**

Students build a computational model tracking the complete environmental life cycle of clothing from fiber production through consumer use to end-of-life disposal.

E **Evaluate Outcomes**

Students run fast fashion default, sustainable wardrobe, and circular fashion scenarios to quantify the environmental difference between disposable and durable clothing systems.

R **Revise & Extend**

Students add labor conditions, microplastic accumulation, or consumer culture pressure to explore the broader social and environmental dimensions of the fast fashion system.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"You're wearing the problem right now. Look at the tag on your shirt — where was it made? What's it made of? How many times will you wear it before it ends up in a landfill?"*

Watch For

Have students check their clothing tags. Record the most common materials and countries of origin. Ask: What do you think the environmental cost of this shirt was?

Slide 2: Learning Objectives

3 min

Say: *"Today we're doing a complete environmental autopsy of a t-shirt — from the cotton field to the landfill — and discovering that the price you paid doesn't include the price the planet pays."*

Watch For

Have students read objectives. Pre-teach 'life cycle assessment' and 'microplastic pollution' with shocking statistics.

Slide 3: Big Question

3 min

Say: *"This shirt costs $5. But making it used 2,700 liters of water, generated 5.5 kg of CO2, and will release 700,000 microplastic fibers every time you wash it. Who pays for that?"*

Watch For

Quick-write: If the environmental cost isn't in the price, where is it? Who or what absorbs that cost? Share out.

Slide 4: LEVER Framework

2 min

Say: *"We're going to model the complete life cycle of clothing — from raw material to landfill — and find the biggest levers for reducing the damage."*

Watch For

Preview LEVER steps. Emphasize that this model tracks multiple environmental impacts simultaneously.

Slide 5: Activity 1: Components

8 min

Say: *"What drives the environmental impact of clothing? Some factors are choices the industry and consumers make, others are physical consequences of those choices."*

Watch For

Guide sorting: Production Volume, Material Type, and Garment Lifespan are external (industry/consumer choices). Water Consumption, Carbon Emissions, and Textile Waste are internal (environmental consequences).

Slide 6: Activity 2: Connections

8 min

Say: *"More production means more water, more carbon, more waste. But lifespan is the hidden variable — if you wear something twice as long, you cut its per-wear impact in half."*

Watch For

Students map connections. Highlight the critical role of garment lifespan as a multiplier/divisor for all impact metrics.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's compare three worlds: fast fashion as usual, a sustainable wardrobe approach, and a circular system where old clothes become new clothes. Which one can the planet survive?"*

Watch For

Students predict annual environmental impact for each scenario for 1,000 consumers. Run simulations. Calculate: How many Olympic swimming pools of water does each scenario use?

Slide 8: Discoveries

5 min

Say: *"The numbers don't lie: wearing clothes longer is the single most powerful thing you can do. Even more than switching materials. And a circular system could cut waste by 80%."*

Watch For

Show real-world fast fashion waste statistics alongside model predictions. Discuss: What would it take to change the industry and consumer behavior?

Slide 9: STEM Challenge

12 min

Say: *"Design a fashion brand that teenagers actually want to wear — but that cuts water use, carbon, and waste by 70%. Prove it's possible with your model data."*

Watch For

Groups design brand strategies specifying materials, durability targets, pricing, and end-of-life programs. Present with environmental impact projections from model.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Production Volume, Material Type, Garment Lifespan
- **INTERNAL:** Water Consumption, Carbon Emissions, Textile Waste Generated

Relationships

- **Production Volume to Carbon Emissions:** POSITIVE (+) — More garments produced means more fiber manufacturing, more factory energy, more chemical processing, and more shipping — each additional garment adds approximately 5-10 kg of CO2 equivalent to the global total. The fashion industry's 10% share of global emissions is driven primarily by the sheer volume of production.
- **Material Type to Water Consumption:** DETERMINES — The choice of fiber fundamentally determines water impact. Conventional cotton requires 2,700 liters per shirt for irrigation alone. Polyester requires less water for production but creates microplastic water pollution during washing. Organic cotton uses 91% less water from irrigation but still requires rainfall. Recycled fibers require 80-90% less water than virgin production.
- **Garment Lifespan to Textile Waste Generated:** NEGATIVE (-) — Longer garment lifespan directly reduces the rate at which clothing enters the waste stream. If the average garment lifespan doubles from 7 to 14 wears, the waste generated per consumer per year is cut approximately in half because fewer replacement garments are needed.

Key Discoveries

- The fast fashion industry's environmental impact is staggering: over 100 billion garments produced annually, 10% of global CO2 emissions, 20% of global water pollution from dyeing, and 92 million tons of textile waste — driven by a business model that incentivizes buying more and wearing less
- Garment lifespan is the single most impactful variable — doubling the number of times a garment is worn before disposal cuts its per-wear environmental footprint in half, regardless of material type
- Polyester produces less water pollution than cotton during production but generates persistent microplastic pollution during washing that cotton does not — every material involves trade-offs, and no option is perfectly clean
- A circular fashion system that recycles textile fiber dramatically reduces both virgin material demand and landfill waste, but current recycling technology can only handle simple, single-fiber garments — blended fabrics (most fast fashion) are nearly impossible to recycle

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Recycling and donating clothes solves the waste problem"	Only 13% of discarded clothing is recycled, and most of that is downcycled into insulation, rags, or carpet padding — not new clothing. Of clothing donated to thrift stores, only 10-20% is sold domestically; the rest is shipped to developing countries where it overwhelms local textile industries and often ends up in landfills anyway. True textile-to-textile recycling is limited by fiber blending — most fast fashion garments contain mixed fibers that cannot be separated for recycling. The model shows that reducing production and extending lifespan have far greater impact than end-of-life recycling.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Natural fibers like cotton are always better for the environment than synthetics"	While cotton avoids the microplastic problem of polyester, conventional cotton farming is enormously water-intensive (2,700 liters per t-shirt), requires heavy pesticide use (16% of global insecticide use for 2.4% of cropland), and degrades soil. Polyester requires less water and land but produces microplastic pollution and relies on petroleum. The model shows that no fiber is universally 'better' — the environmental profile depends on which impacts you prioritize. Organic cotton, recycled polyester, and hemp each represent different trade-off profiles.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Individual consumer choices can fix the fast fashion problem"	While consumer choices matter — buying less, wearing longer, choosing better materials — the scale of the problem is fundamentally driven by industry business models. Fast fashion companies produce 100+ billion garments per year because the business model profits from disposability and trend cycling. Systemic change requires policy (extended producer responsibility, environmental labeling), technology (textile recycling at scale), and business model transformation (rental, resale, repair services) alongside consumer behavior change.